

WEB AND SOCIAL MEDIA SEARCH AND ANALYSIS

Machine Translation Discourse on Bluesky

Tool, threat, or both? What 3,864 posts and their social network reveal

Linda Stalidzane • Pelinsu Topaloglu

Bachelor of Science in Artificial Intelligence — Prof. Marco Viviani

Tool, threat, or both?



DeepL, Google Translate, and LLM-based translation are now used by casual users, professional translators, and developers alike. We studied how Bluesky talks about it — and whether professionals and the general public frame it differently.

RQ1

Sentiment & framing

Tool, threat, or both?

RQ2

Professional impact

How do translators discuss career effects?

RQ3

Utility vs. displacement

Do people hold both views at once?

RQ4

Network structure

Do professionals and the public form separate communities?

Collecting a year of MT discourse



Bluesky API (atproto), 12-month window: seed search on MT keywords/hashtags → full reply-thread crawl → follower crawl + profile fetch for bio-based labeling.



Author type, by bio keyword match:



“unknown” = profile fetched, no bio — kept separate from “general” deliberately

Content analysis: two tools per task, on purpose

For sentiment and emotion, we deliberately used a lexicon-based tool and a transformer, and cross-checked them — this surfaced real weaknesses in the lexicon tools rather than hiding them.

Sentiment

VADER (lexicon) vs.
cardiffnlp RoBERTa (transformer)

Emotion

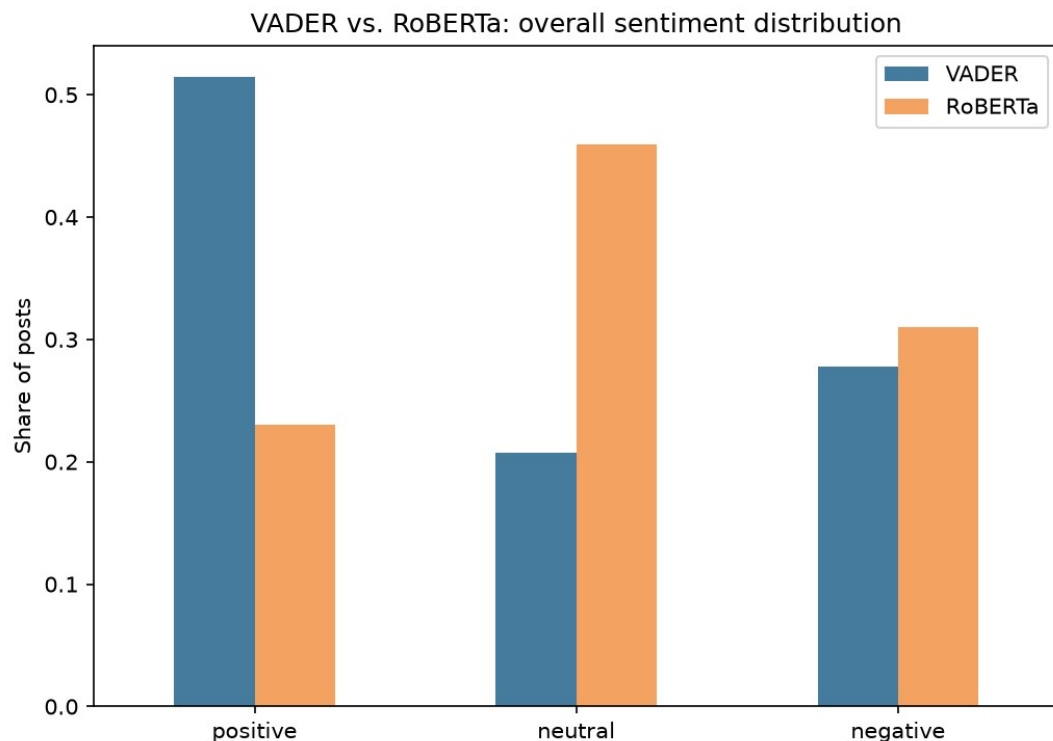
NRCLex (dictionary) vs.
j-hartmann DistilRoBERTa (transformer)

Entities

spaCy NER + curated list of
known MT tools/companies

Why compare? Because “trust me, it's positive” isn't good enough science — see next slides.

Sentiment: the tools disagree — a lot



54.1% agreement

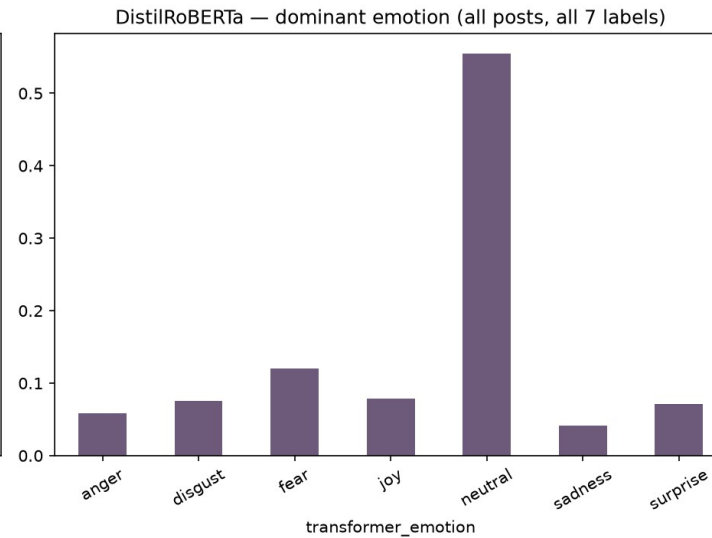
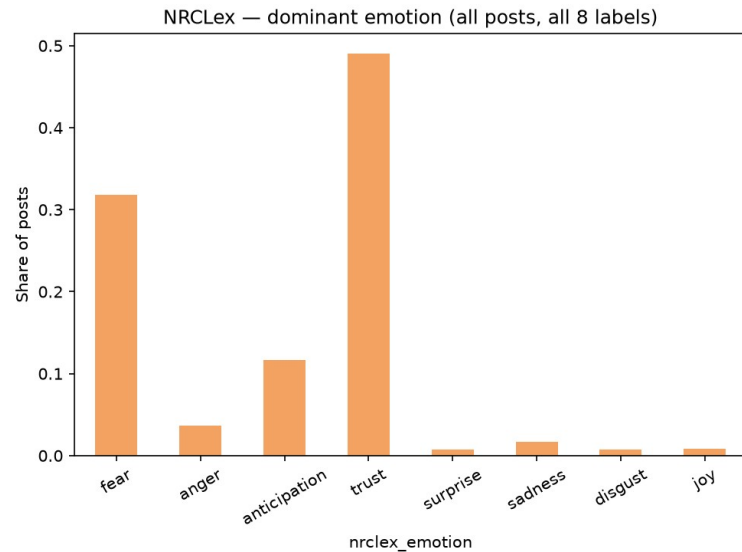
between VADER and RoBERTa — 458 posts (11.9%) in outright opposite-polarity disagreement.

VADER: fooled by individually-positive words.

“Perfect example of how machine translation is totally untrustworthy” → VADER: +0.57 positive (from “perfect”). RoBERTa: negative — correctly.

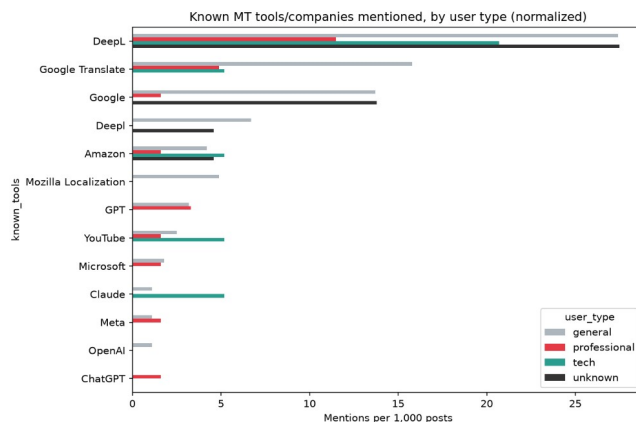
Takeaway: the discourse is genuinely mixed (23%/46%/31%), not predominantly positive as VADER alone suggests. Professionals score most positive on average (mean 0.201 vs. 0.139 general) — consistent with acknowledging MT's day-to-day utility, not framing it purely as a threat.

When the tool matters more than the topic



NRClex says “trust” or “fear” on ~80% of posts — almost regardless of content. Conditioned on posts where DistilRoBERTa confidently detects a specific emotion, the two tools still agree only 15.6% of the time.

Why: “trust” is an oversized NRC lexicon category that fires on ordinary vocabulary — not genuine signal.



Entities: 5,523 extracted. DeepL & Google Translate dominate for every group; tech users lean toward Claude/LLM-based translation at a disproportionate rate.

Network analysis: who talks to whom



Follow graph and reply graph kept separate (different social relations). Structural analysis restricted to the post-authors subgraph — the full follow graph is an ego-network sample, capped per account by API pagination.

Centrality

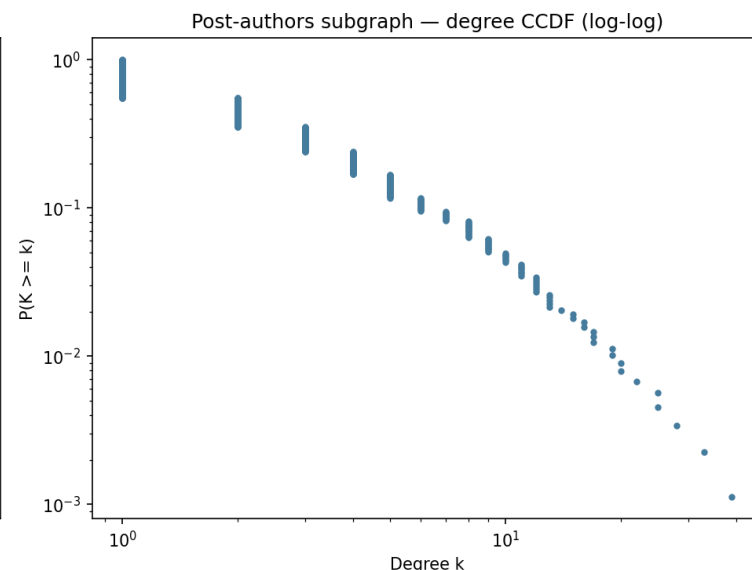
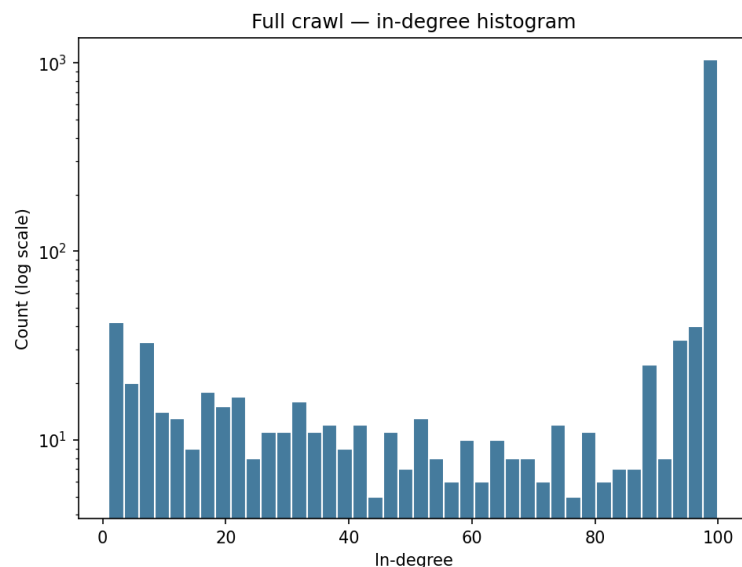
In/out-degree, PageRank, betweenness, closeness — who's visible, who bridges groups

Community detection

Louvain, Greedy Modularity, FluidC, Girvan–Newman — compared on the same graph

Reproducibility note: Louvain/FluidC results shift slightly run-to-run (Python hash randomization) — acknowledged, not hidden.

A loosely connected discourse



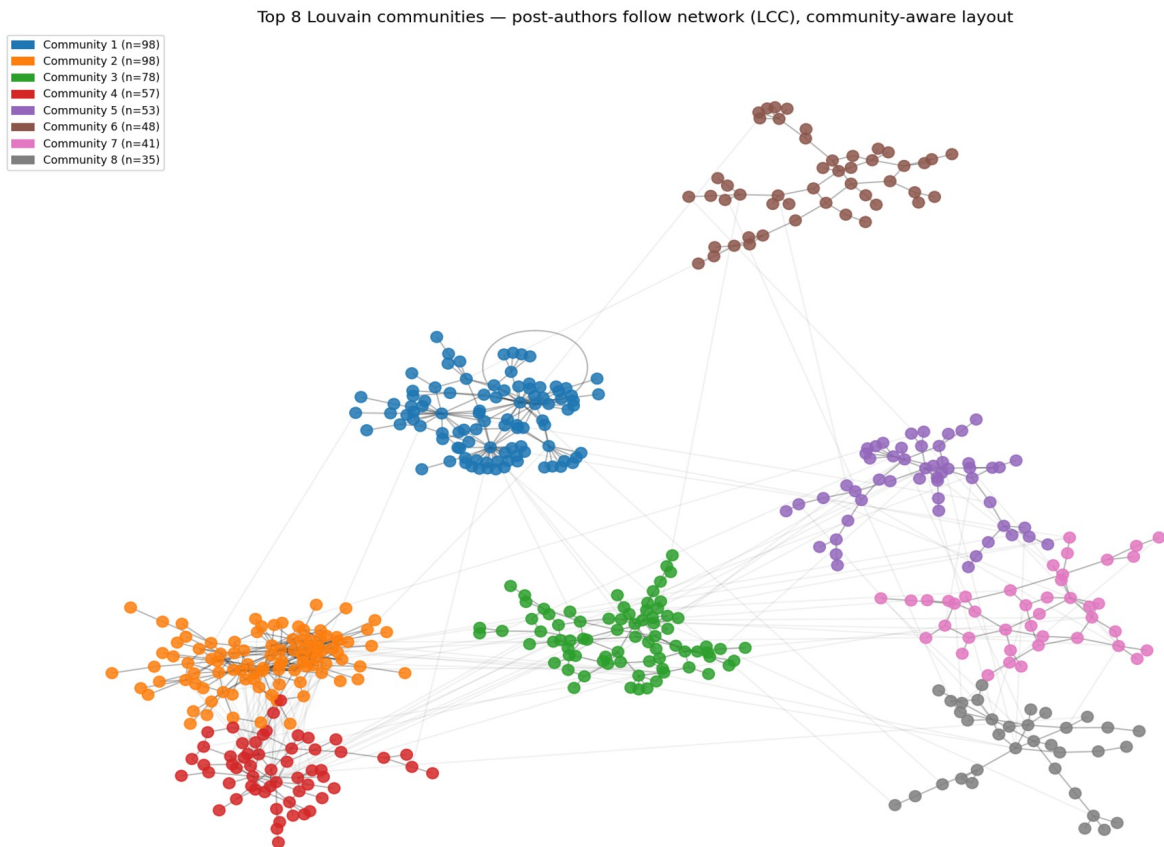
1,339 nodes, 1,320 edges (density 0.0007), 535 fragments — largest component covers only 676 nodes (~half). Low reciprocity (0.16): attention flows one way.

Spike at in-degree ≈ 100 = API pagination cap, a collection artifact, not real structure.

Follow attention concentrates on professionals: 13 of top 15 by in-degree, 10 of top 15 by closeness. Betweenness splits exactly 5/5 — professionals bridge, but don't monopolize.

The reply network inverts this: most-replied accounts are mostly general users — one viral thread (364 replies, $\sim 5.6\times$ the runner-up) from far outside the translation community. Professionals are followed; virality happens around general accounts.

Community detection: not one hub, a few



16 communities found (Louvain, modularity 0.728, consistent with 3 other algorithms). Two are professional-majority — not the single hub we expected.

98 nodes — 134 professional vs. 112 general
MTPE rates, working conditions, industry news

57 nodes — 57 professional vs. 49 general
Narrower margin, more heterogeneous

The rest: general-dominated, organized by fandom/topic (game localization, fan translation) — not by profession.

RQ4: “professionals vs. the public” is too simple — it's a few professional hubs surrounded by many independent fan conversations.

OVERALL CONCLUSION

Tools and products, not tribes

- Professional translators are the most visible, most-followed, and — most positive on average: they discuss MT with nuance, acknowledging utility while contesting working conditions.
- General users don't oppose them as a bloc — they scatter across many product- and fandom-centered conversations (game localization, fan translation), each triggered by a specific encounter with MT.
- Conversational attention is event-driven (one viral thread dominates replies), while follow-attention is sustained and professional-led — two different kinds of “importance.”
- DeepL and Google Translate dominate the conversation for every group: the discourse organizes around concrete tools, not an abstract pro/anti divide.

What this analysis can't claim



- Single platform, one ~12-month window — findings may not generalize.
- Keyword-based topical filter and author labeling, not validated classifiers.
- Generic NER (spaCy) mis-tags domain acronyms (MTPE, LLM) as organizations.
- Lexicon tools (VADER, NRCLEX) diverge substantially from transformers — no gold-standard annotation to say which is “correct,” only that they disagree.
- Ego-network sample with a 100-follower cap per account; Louvain/FluidC community results vary slightly run-to-run.

Thank you

Questions?

Linda Stalidzane • Pelinsu Topaloglu